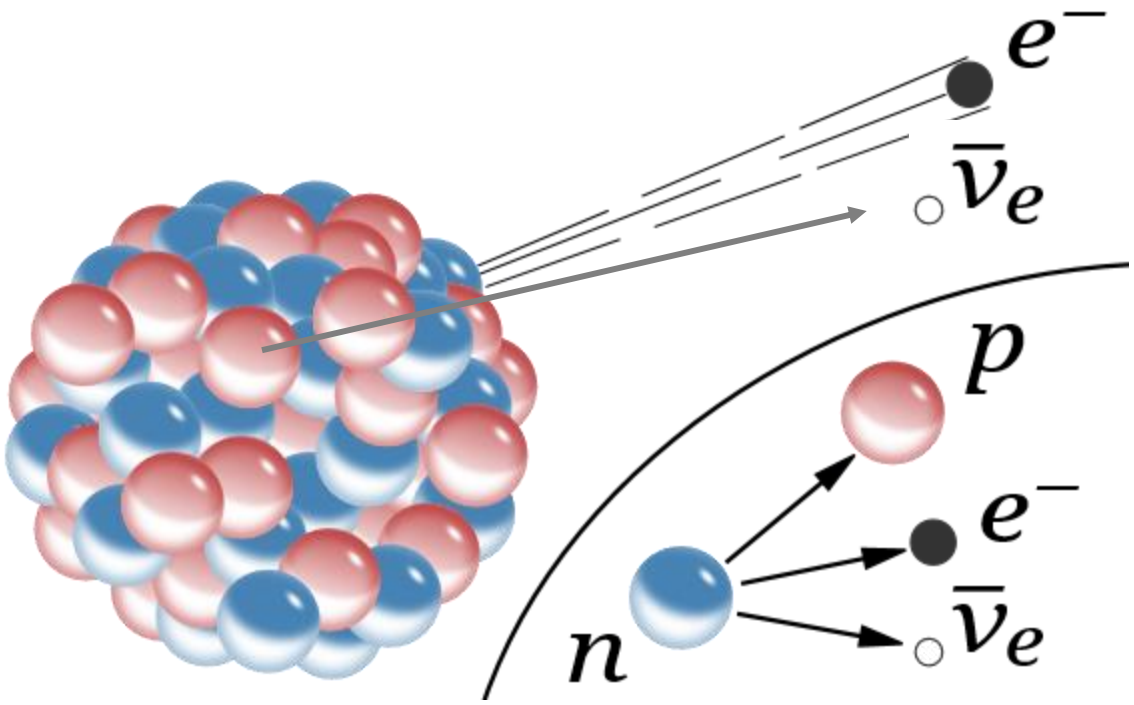
SH1011 Modern fysik

Föreläsning 13:
Repetition
Hur fungerar MR?
Alfasönderfall.
Betasönderfall.
Gammasönderfall.
Naturlig radioaktivitet.
Kärnreaktioner.
Datering med hjälp av radioaktivitet.
Biologiska effekter av strålning.
Fission.
Kärnreaktorer.
Fusion.
(PET och SPECT)



Mats Danielsson

Professor, Institutionen för fysik, KTH

mats.danielsson@mi.physics.kth.se

Hur fungerar MRI?

Protoner och neutroner har spinn ½…

… och därmed ett magnetiskt dipolmoment

$$\boldsymbol{\mu}_S = g\frac{e}{2m}\boldsymbol{S}$$

Gyromagnetisk kvot γ

Mäts ofta i multipler av

$$\mu_N = \frac{e\hbar}{2m_p}$$

Nuklearmagnetonen

Partikel	g
Elektron	-2.002 319 304 362 56(35)
Proton	5.585 694 6893(16)
Neutron	-3.826 085 45(90)

Spin

$$S = \sqrt{s(s+1)}\hbar = \sqrt{\tfrac{1}{2}(\tfrac{1}{2}+1)}\hbar \quad S_z = \pm\tfrac{1}{2}\hbar$$

Zeeman effect

$$U = -\mu_z B = m_l \frac{e\hbar}{2m} B = m_l \mu_B B$$

ROYAL INSTITUTE OF TECHNOLOGY

Kärn Magnetism

- Kärnor med udda antal neutroner eller protoner har magnestiskt moment.
- Kärnor med jämt antal neutroner och protoner har inget magnetiskt moment och kan inte observeras med MR

ROYAL INSTITUTE OF TECHNOLOGY

Kärn magnetism forts

- Viktiga atomkärnor utan magnetiskt moment:

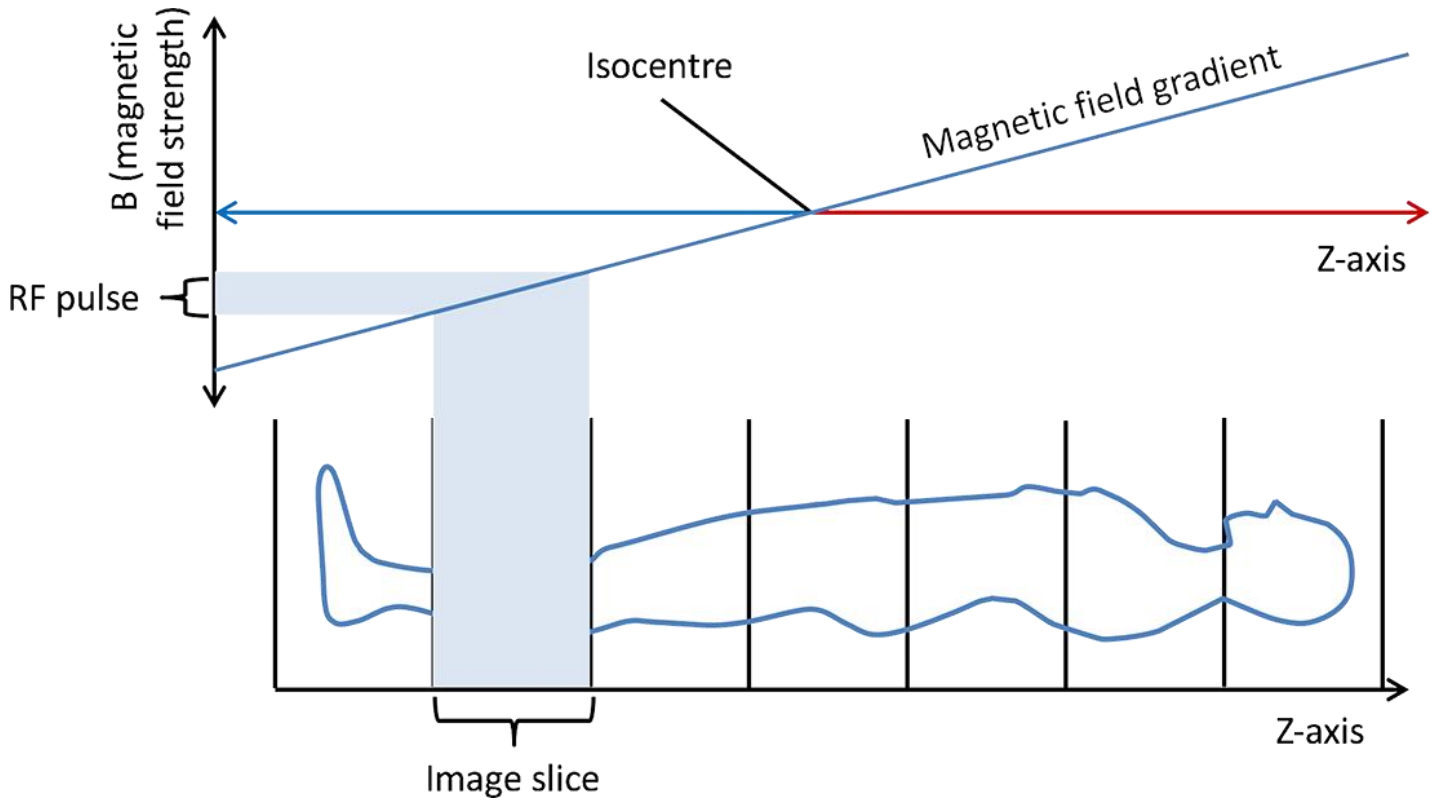
  $^{16}_{8}\mathbf{O}$ ==> 8 neutrons, 8 protons, 8 electrons

  $^{12}_{6}\mathbf{C}$ ==> 6 neutrons, 6 protons, 6 electrons

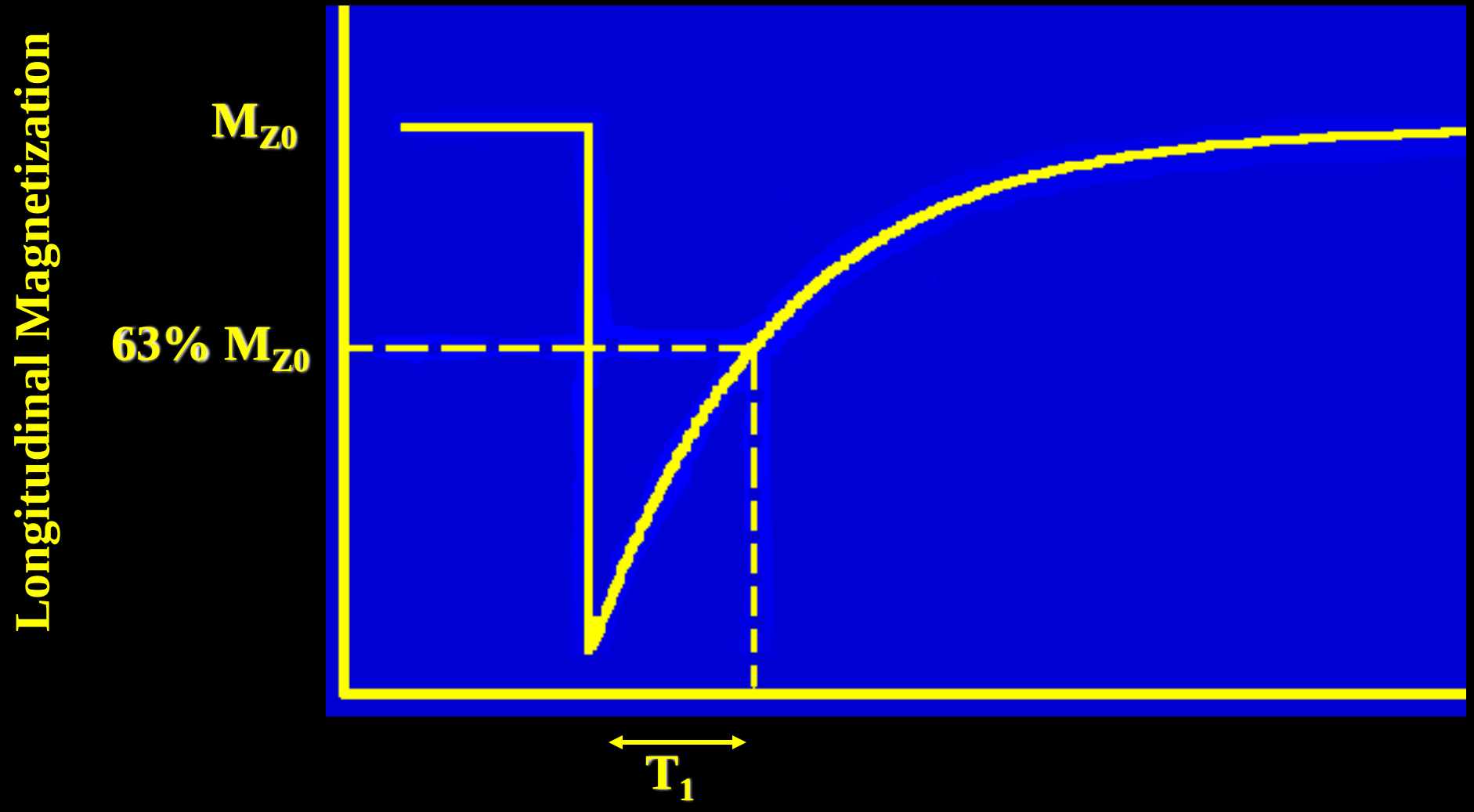
- Väte förekommer rikligt (H_2O, CH_2:fat) och har det största magnetiska momentet.

$^{1}_{1}\mathbf{H}$ ==> 1 proton, 1 electron

Med en magnetfältsgradient kan man välja skiva att avbilda



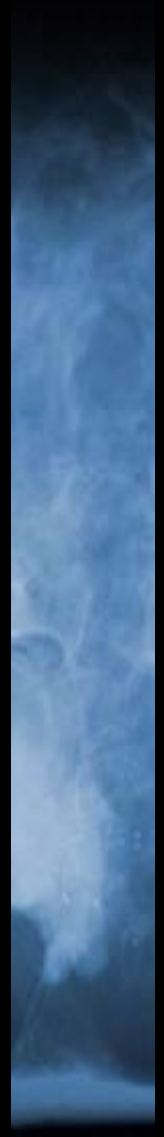
By Sarah Abdulla, https://www.radiologycafe.com/radiology-trainees/frcr-physics-notes


Longitudinal Magnetization
M_{Z0}
63% M_{Z0}
T_1
Time

ROYAL INSTITUTE OF TECHNOLOGY

Approximate T_1 and T_2 Values for Human Tissue (37 ^{o}C)

Tissue	T_1 at 1.5 T (msec)	T_1 at 0.5 T (msec)	T_2 (msec)
Skeletal Muscle	870	600	47
Liver	490	323	43
Kidney	650	449	58
Spleen	780	554	62
Fat	260	215	84
Gray Matter	920	656	101
White Matter	790	539	92
Cerebrospinal Fluid	>4,000	>4,000	>2,000



Gadolinium

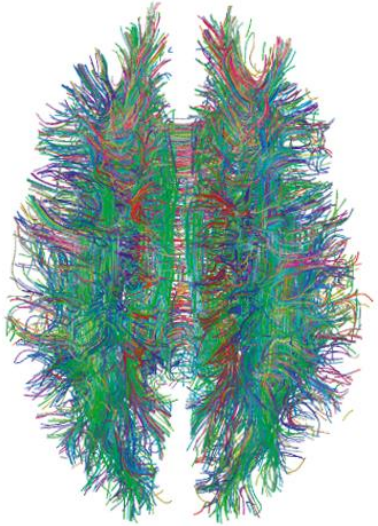
Standard kontrastmedel vid MR

Z=64 så man ser detta även med CT
K-alfa vid 43 keV

Magnetresonansavbildning: tillämpningar

Cancer, mjukvävnad…

Diffusionstensoravbildning av vit hjärnsubstans



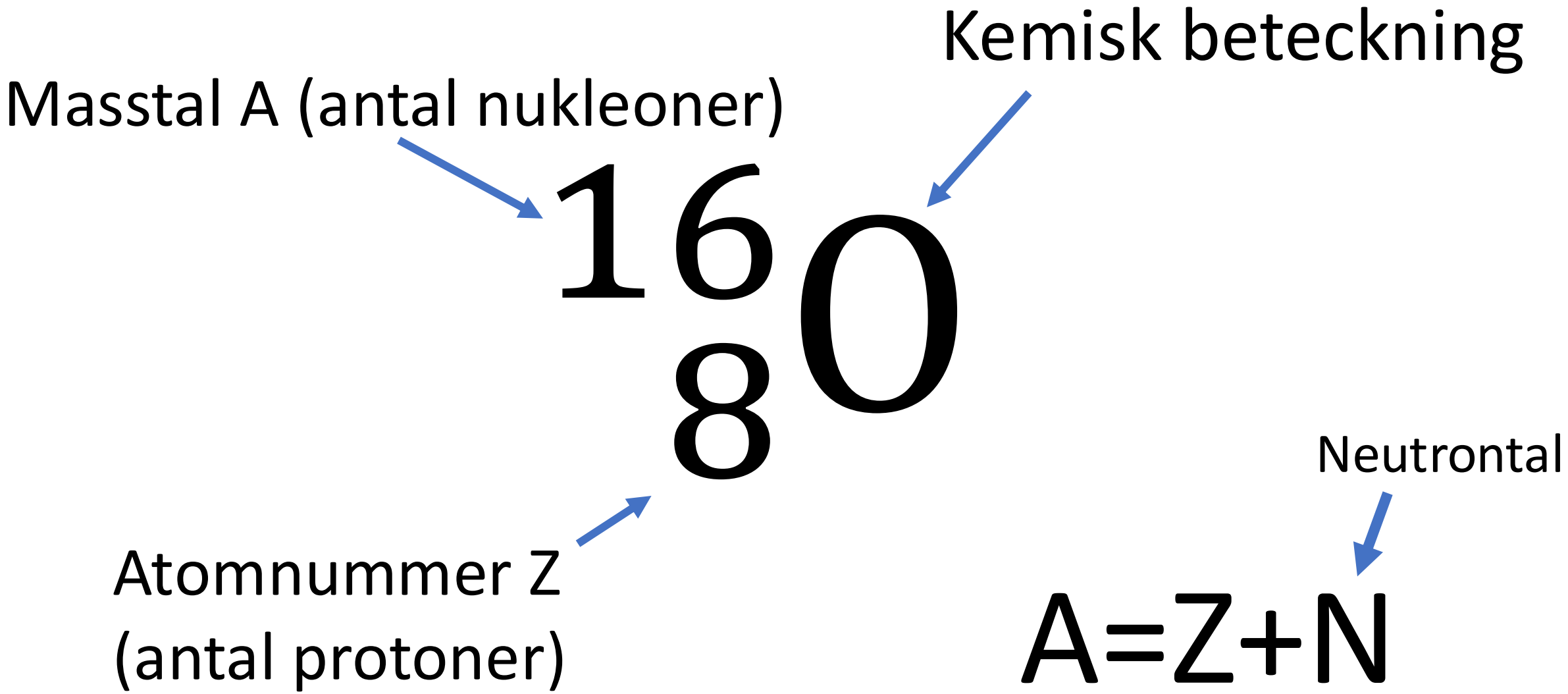
Funktionell MRI (fMRI)



Respons vid uppläsning av ord

Xavier Gigandet
https://commons.wikimedia.org/wiki/File:White_Matter_Connections_Obtained_with_MRI_Tractography.png

Beteckning för en nuklid



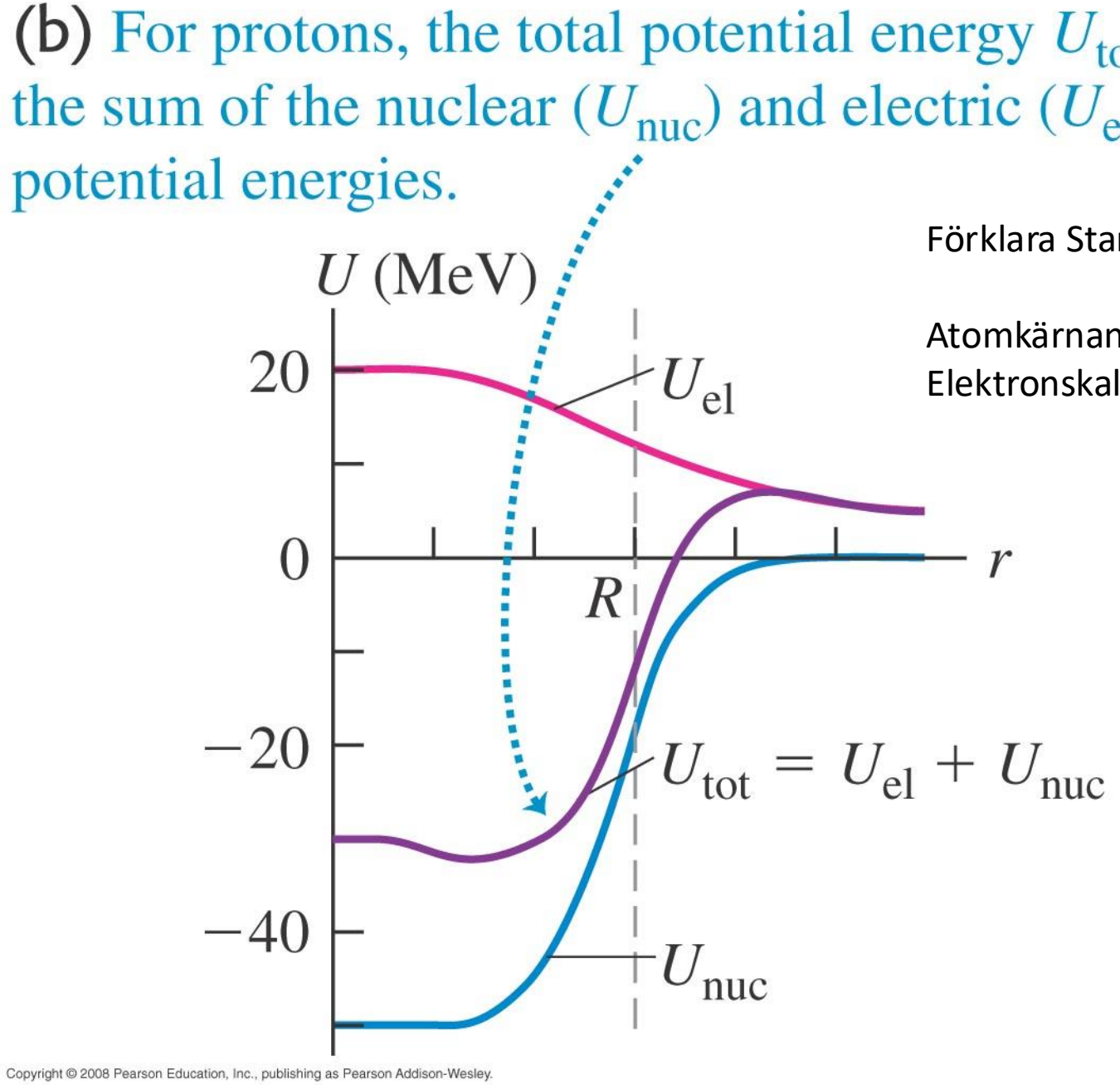
Massenheten 1 u

1 u = 1/12 av massan av en atom av kolisotopen ^{12}C

1 u = $1{,}660539067 \cdot 10^{-27}$ kg
1 u = 931.46 MeV/c2

-protonens massa 938.272 013 MeV/c2
-neutronens massa 939.565560 MeV/c2

Figure 43.3b



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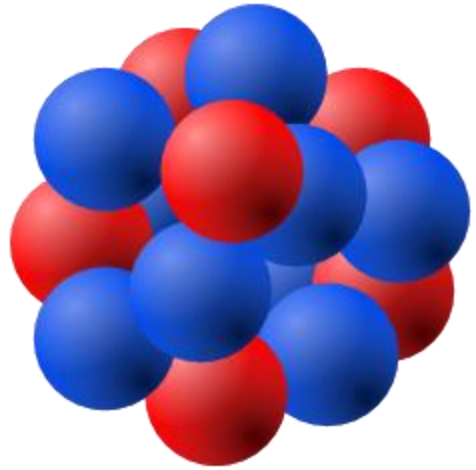
Förklara Stark växelverkan - som binder atomkärnan

Atomkärnan har en skalstruktur som påminner om Elektronskalen i en atom

Kärnradien

$$R = R_0 A^{\frac{1}{3}}$$

Masstal (antal nukleoner)



$$R_0 = 1.2 \cdot 10^{-15}\ m$$

Bindningsenergi

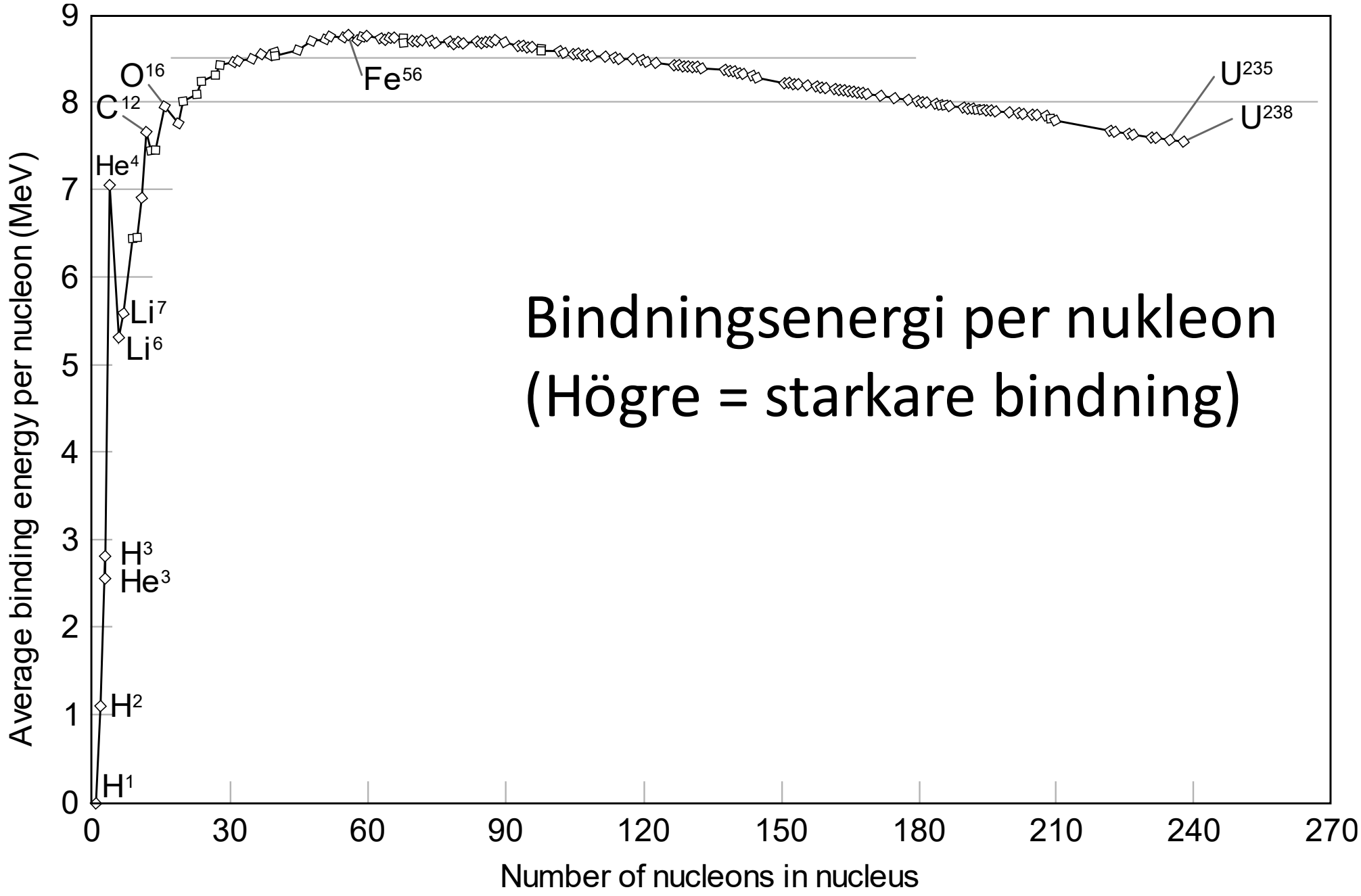
Den energi som måste tillföras för att dela upp en nuklid i Z väteatomer och N neutroner

Väteatomens massa

Neutronens massa

$$E_B = \left(Z M_H + N m_n - {}^{A}_{Z}M\right)c^2$$

Nuklidens massa



Bindningsenergi per nukleon
(Högre = starkare bindning)

FastFission https://en.wikipedia.org/wiki/Nuclear_binding_energy#/media/File:Binding_energy_curve_-_common_isotopes.svg

Sönderfall, antal kärnor som är kvar

$$N(t) = N_0 e^{-\lambda t}$$

$$T_{\text{mean}} = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2} = \frac{T_{1/2}}{0.693}$$

(lifetime T_{mean}, decay constant λ, and half-life $T_{1/2}$)

Vad kan hända en atomkärna

-alfasöndefall

-betasönderfall
tre olika

-Exiterat tillstånd går tillbaka till grundtillståndet genom emission av gamma strålning

-Fission

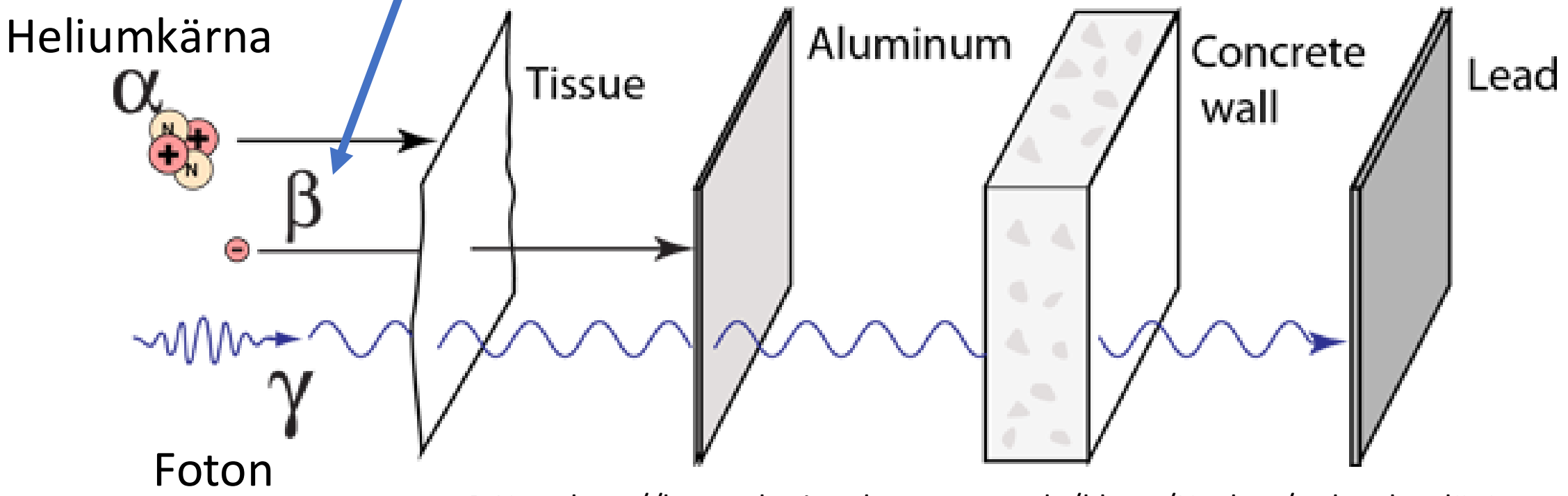
-Fusion

Typer av radioaktiv strålning

1 Gy = 1 J/kg absorberad strålningsenergi

1 Sv = 1 Gy x biologisk viktfaktor Q som beskriver farligheten
Gamma och beta Q=1, Alpha Q= 20 (10 mikrometer vävnad)

Elektron (β^-)
eller positron (β^+)



R.Nave http://hyperphysics.phy-astr.gsu.edu/hbase/Nuclear/radact.html#c1

Mått på stråldos till levande organismer

Absorberad dos

$$D = \frac{\text{absorberad energi}}{\text{massa}}$$

Enhet :1 Gray $(\text{Gy}) = 1\,\frac{\text{J}}{\text{kg}}$

Korrigeras för strålningens "farlighet" med factor Q

Ekvivalent dos

$$H = QD$$

Ekvivalent Dos

Enhet :1 Sievert $(Sv)1\,\frac{\text{J}}{\text{kg}}$

Korrigeras i sin tur för vävnadens känslighet med en factor D

Effektiv dos

$$D_{\text{effective}} = \sum_{\text{Tissues}} w_T H_T$$

Enhet :1 Sievert $(Sv) = 1\,\frac{\text{J}}{\text{kg}}$

Båda mäts i Sievert (Sv) = J/kg

Ekvivalent dos

$$H = QD$$

Effektiv dos

$$D_{\text{effective}} = \sum_{\text{Tissues}} w_T H_T$$

Table 2. Recommended radiation weighting factors. (ICRP 103)

Radiation type	**Radiation weighting factor, w_R**
Photons	1
Electrons[a] and muons	1
Protons and charged pions	2
Alpha particles, fission fragments, heavy ions	20
Neutrons	A continuous function of neutron energy

=RBE, relative biological effectiveness

Boken ger ett annat värde för protoner!

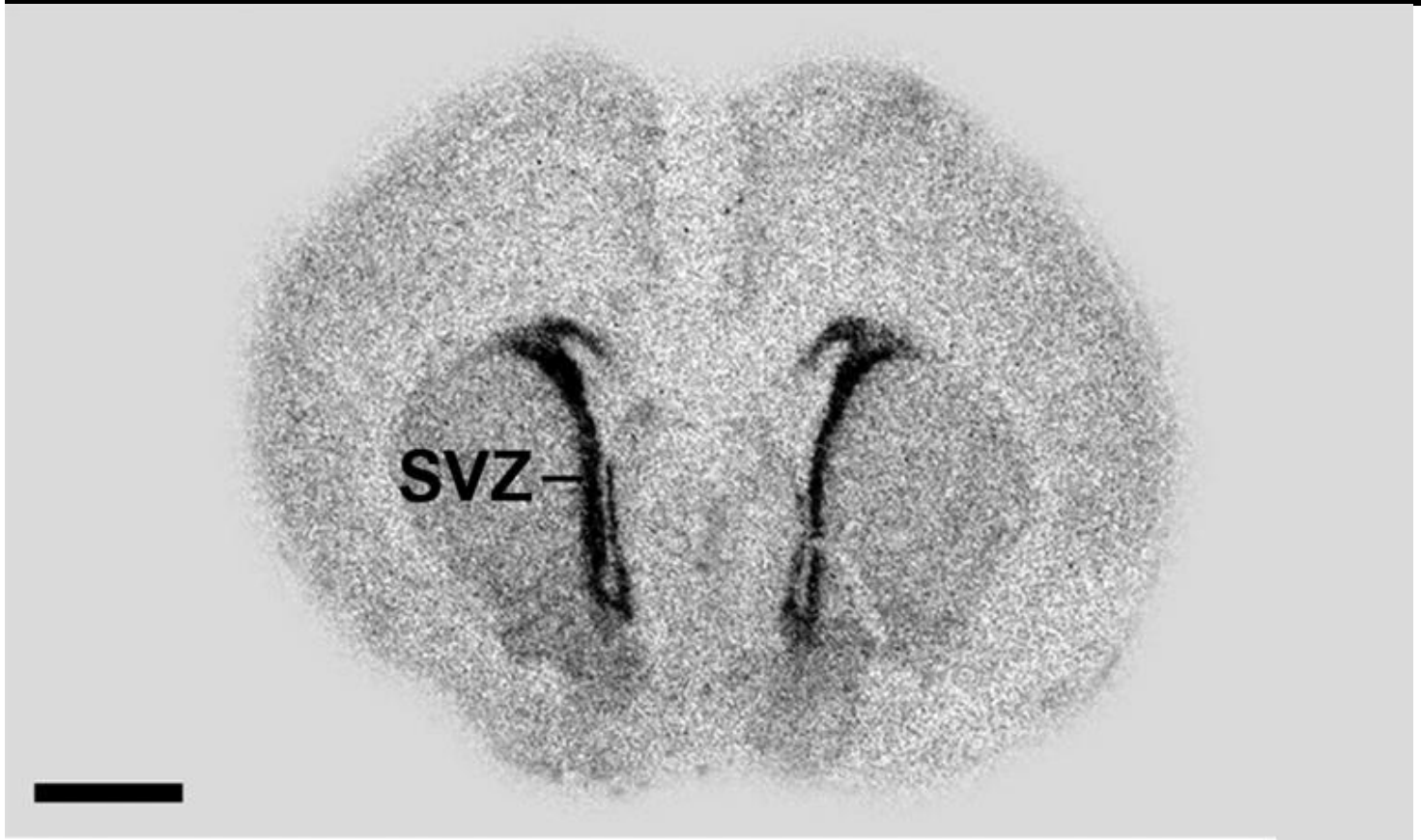
Table 3. Recommended tissue weighting factors.

Tissue	w_T	$\sum w_T$
Bone-marrow (red), Colon, Lung, Stomach, Breast, Remainder tissues*	0.12	0.72
Gonads	0.08	0.08
Bladder, Oesophagus, Liver, Thyroid	0.04	0.16
Bone surface, Brain, Salivary glands, Skin	0.01	0.04
	Total	1.00

* Remainder tissues: Adrenals, Extrathoracic (ET) region, Gall bladder, Heart, Kidneys, Lymphatic nodes, Muscle, Oral mucosa, Pancreas, Prostate (♂), Small intestine, Spleen, Thymus, Uterus/cervix (♀).

Prince, Links, Medical Imaging Signals and systems

Autoradiografi (^{3}H eller ^{125}I)



((Bloomberg) -- Pfizer Inc.'s shares fell 11 percent, wiping out $21 billion of market value, after the world's biggest drugmaker ended development of its most important new treatment, a cholesterol medicine designed to replace Lipitor when its patent expires.Pfizer ended studies of the cholesterol pill torcetrapib Dec. 2 because deaths among patients taking it were 60 percent higher than in a group who didn't get the drug. The stock plunged $2.96 to $24.90 at the close of New York Stock Exchange composite trading. The shares earlier touched $23.50.

Alfa sönderfall

$$^{A}_{Z}X \rightarrow ^{A-4}_{Z-2}Y + ^{4}_{2}He$$

Exempel Astat

$$^{211}_{85}At \rightarrow ^{207}_{83}Bi + ^{4}_{2}He$$

Exempel Radon

$$^{222}_{86}Rn \rightarrow ^{218}_{84}Po + ^{4}_{2}He$$

Beta sönderfall

Beta minus: $n \rightarrow p^+ + e^- + \bar{\nu}_e$

Ex: $^{14}_{6}C \rightarrow ^{14}_{7}N + e^- + \bar{\nu}_e$

Beta plus: $p^+ \rightarrow n + e^+ + \nu_e$

Ex: $^{18}_{9}F \rightarrow ^{18}_{8}O + e^+ + \nu_e$ "FDG"

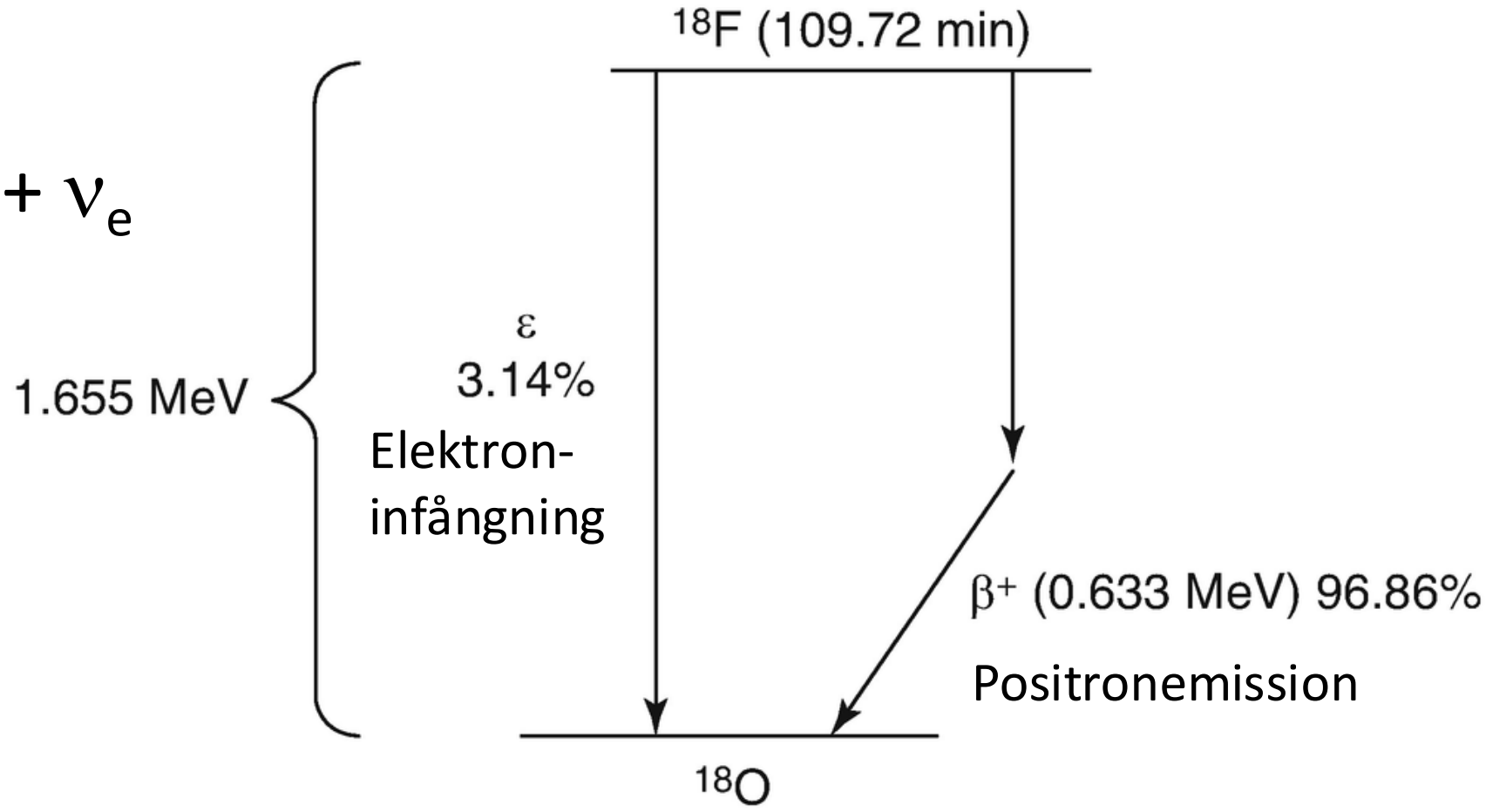
Electron Capture (EC):

$p^+ + e^- \rightarrow n + \nu_e$

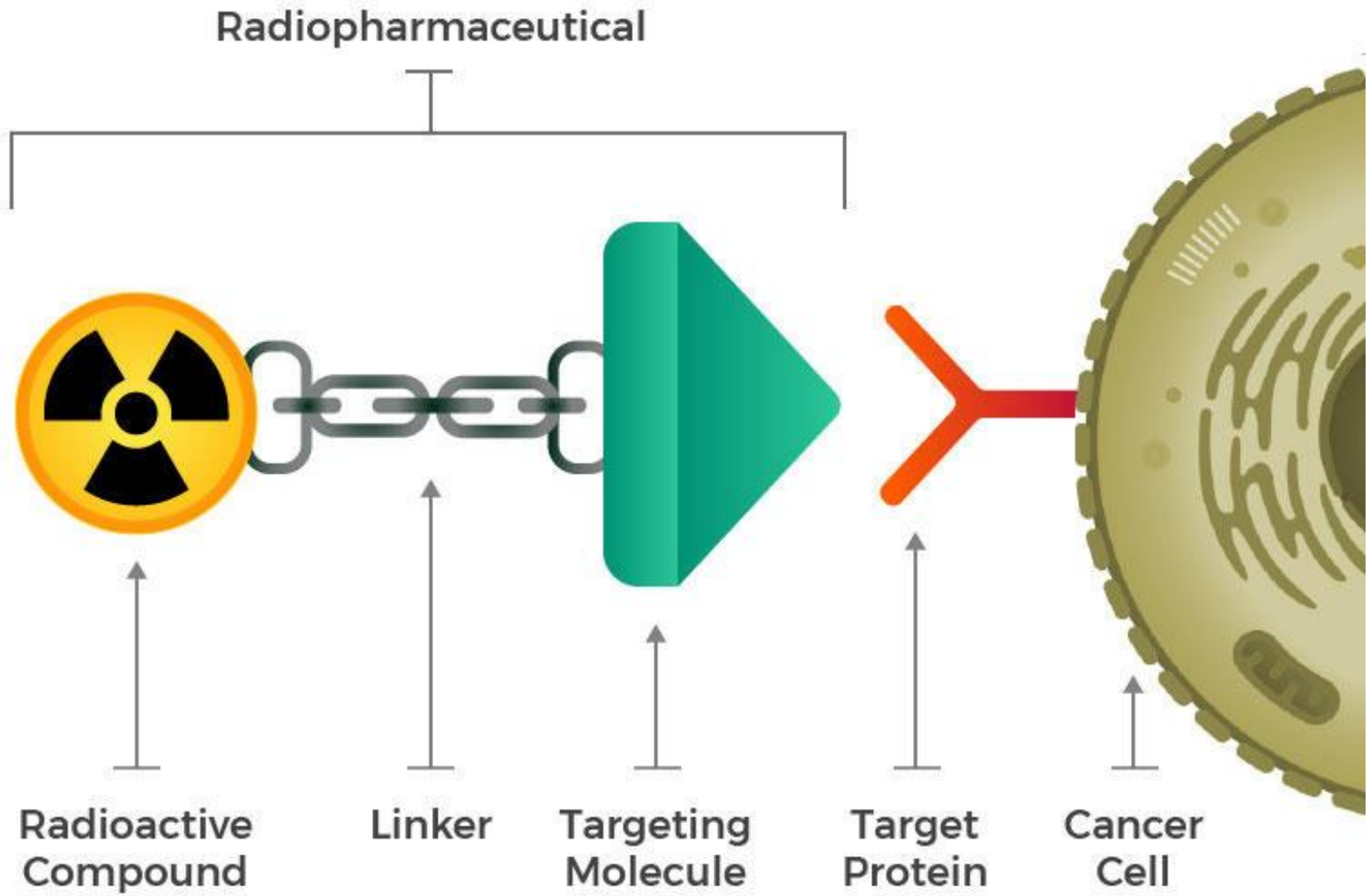
Ex: $^{186}_{75}Re + e^- \rightarrow ^{186}_{74}W + \nu_e$ Rhenium

^{18}F – En vanlig radioaktiv nuklid för PET

$$^{18}_{9}F \rightarrow ^{18}_{8}O + e^{+} + \nu_e$$

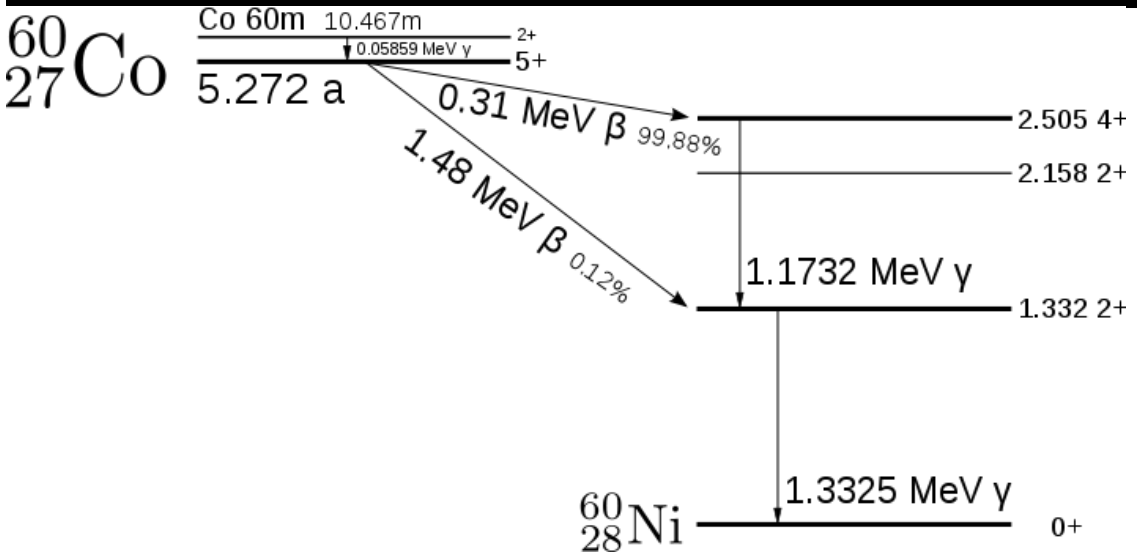


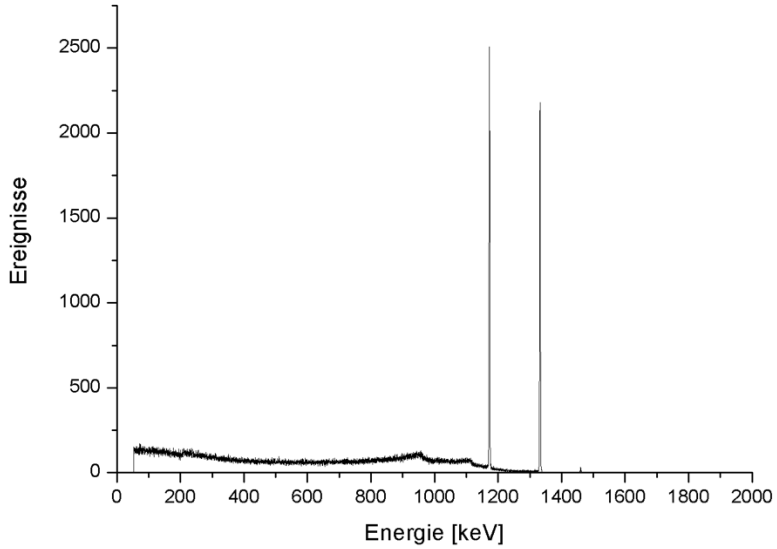
Ermert, Neumaier, Radiopharmaceutical chemistry pp 273-283


Radiopharmaceutical
Radioactive Compound
Linker
Targeting Molecule
Target Protein
Cancer Cell

Gammasönderfall

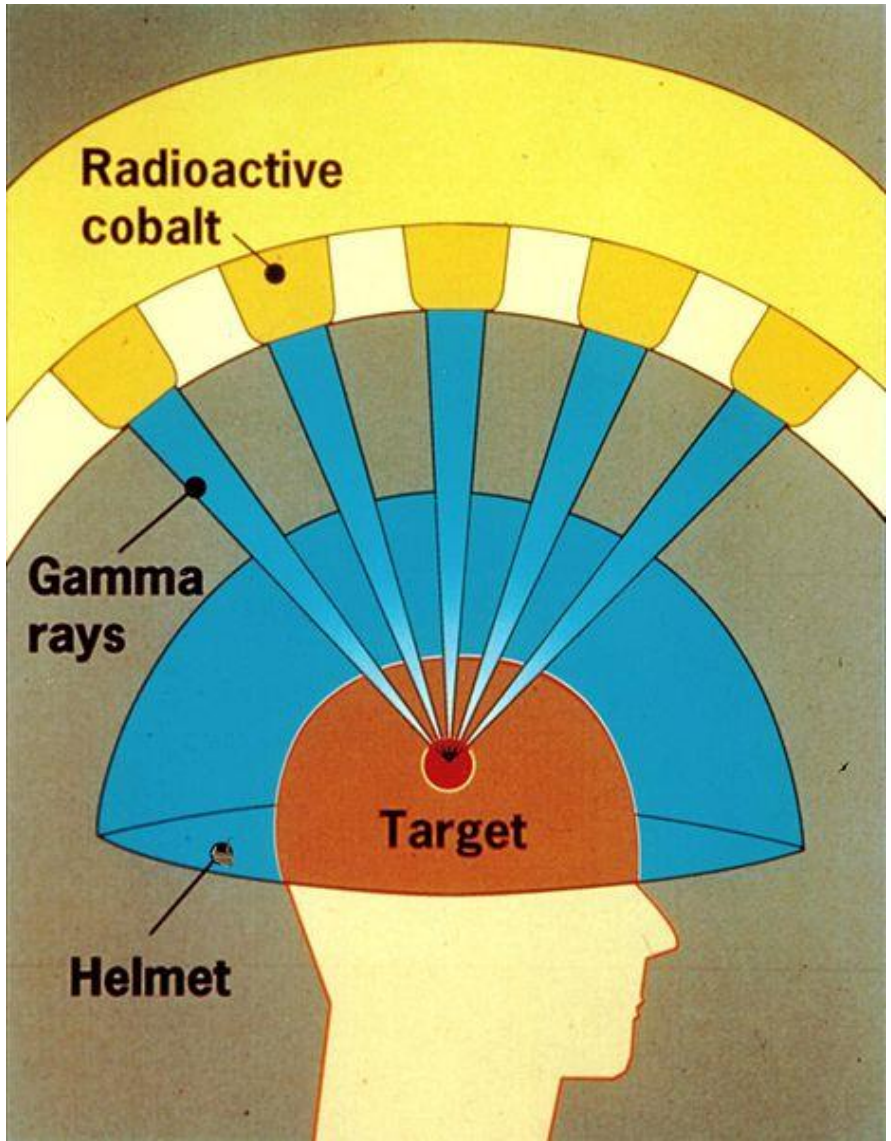
Exempel gamma sönderfall





Hur ändras A och Z vid γ sönderfall?

Exempel Gammakniven



Prof. Leksell

Elekta AB



$$Q = (m_{initial} - m_{final})\,c^2$$

$Q > 0$ innebär att energi frigörs

$Q < 0$ innebär att energi måste tillföras

Ex $^{18}_{9}F \rightarrow {}^{18}_{8}O + e^+ + \nu_e$

Isotopmassor inkluderar elektroner

$m_F = 18.0009380\ u \quad m_O = 17.999160 \quad m_{e^+} = 0.0005485\ u$

OBSERVERA

$m_{\nu e} \simeq 0$

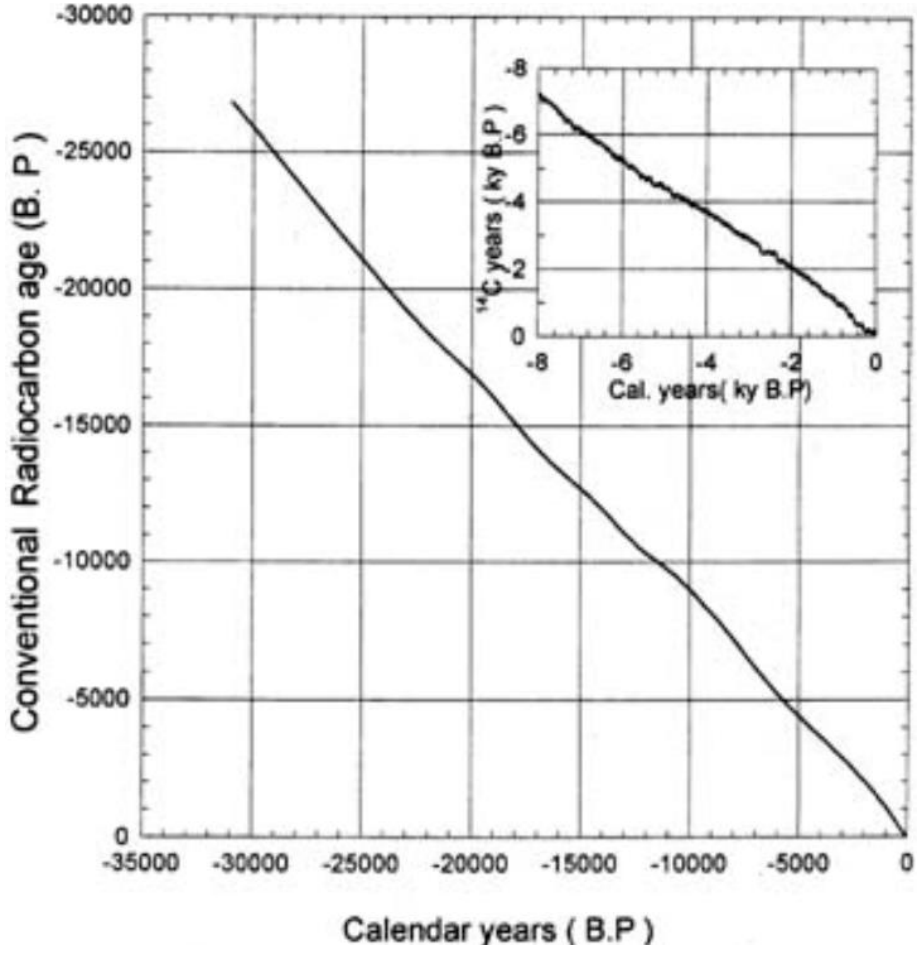
$$Q = (m_F - (m_O + m_{e^+} + m_{e^-}))\,c^2 = 0.0006798\ u\cdot c^2$$

$1\ u = 931.5\ MeV/c^2 \Rightarrow Q = 0.634\ MeV$

C-14-datering

- ^{14}C sönderfaller till ^{14}N med en halveringstid på 5730 år
- Skapas genom att kosmiska strålar reagerar med kväve i atmosfären
- Tas upp av organismer så länge de lever

Carbon-14 dating and other applications in earth sciences
D. Lal S. Krishnaswami



Variationer i koncentration
kalibreras med dendrokronologi (årsringar)

Datering av mineral

Halveringstiden för Uran-238 är cirka 4,468 miljarder år

Skanderna ca 500 miljoner år

Alperna och Himalaya ca 50 miljoner år

1. $^{238}_{92}U \rightarrow {}^{234}_{90}Th + \alpha$ (Half-life: 4.468 billion years)
2. $^{234}_{90}Th \rightarrow {}^{234}_{91}Pa + \beta^- + \bar{\nu}e$ (Half-life: 24.1 days)
3. $^{234}_{91}Pa \rightarrow {}^{234}_{92}U + \beta^- + \bar{\nu}e$ (Half-life: 6.7 hours)
4. $^{234}_{92}U \rightarrow {}^{230}_{90}Th + \alpha$ (Half-life: 245,500 years)
5. $^{230}_{90}Th \rightarrow {}^{226}_{88}Ra + \alpha$ (Half-life: 75,380 years)
6. $^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + \alpha$ (Half-life: 1,600 years)
7. $^{222}_{86}Rn \rightarrow {}^{218}_{84}Po + \alpha$ (Half-life: 3.8 days)
8. $^{218}_{84}Po \rightarrow {}^{214}_{82}Pb + \alpha$ (Half-life: 3.1 minutes)
9. $^{214}_{82}Pb \rightarrow {}^{214}_{83}Bi + \beta^- + \bar{\nu}e$ (Half-life: 26.8 minutes)
10. $^{214}_{83}Bi \rightarrow {}^{214}_{84}Po + \beta^- + \bar{\nu}e$ (Half-life: 19.9 minutes)
11. $^{214}_{84}Po \rightarrow {}^{210}_{82}Pb + \alpha$ (Half-life: 164 microseconds)
12. $^{210}_{82}Pb \rightarrow {}^{210}_{83}Bi + \beta^- + \bar{\nu}e$ (Half-life: 22.3 years)
13. $^{210}_{83}Bi \rightarrow {}^{210}_{84}Po + \beta^- + \bar{\nu}e$ (Half-life: 5 days)
14. $^{210}_{84}Po \rightarrow {}^{206}_{82}Pb + \alpha$ (Stable) (Half-life: 138.4 days)

Fusion

JET, Iter, Novatron

$^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + {}^{1}_{0}n + 17.6\ MeV$

Solen

Step 1: Proton-Proton Fusion

Two protons (^{1}H) fuse to form deuterium (^{2}H), emitting a positron (e^{+}) and a neutrino (ν_e), along with energy:

$^{1}H + {}^{1}H \rightarrow {}^{2}H + e^{+} + \nu_e + 0.42\ MeV$

Step 2: Deuterium-Proton Fusion

The deuterium nucleus then fuses with another proton, forming helium-3 (^{3}He) and releasing a gamma ray (γ):

$^{2}H + {}^{1}H \rightarrow {}^{3}He + \gamma + 5.49\ MeV$

Step 3: Helium-3 Fusion

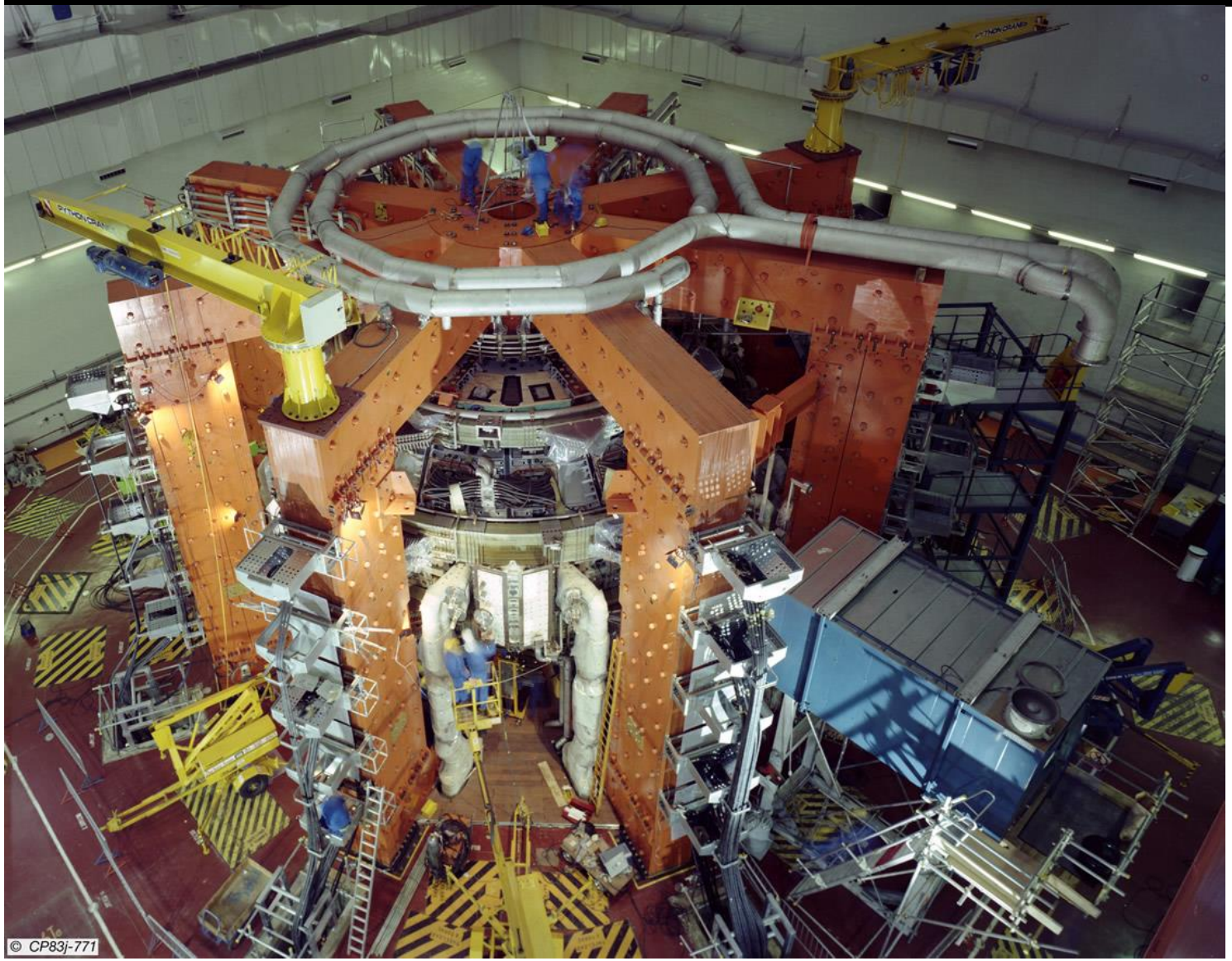
Two helium-3 nuclei fuse to form helium-4 (^{4}He) and two protons:

$^{3}He + {}^{3}He \rightarrow {}^{4}He + 2\ {}^{1}H + 12.86\ MeV$

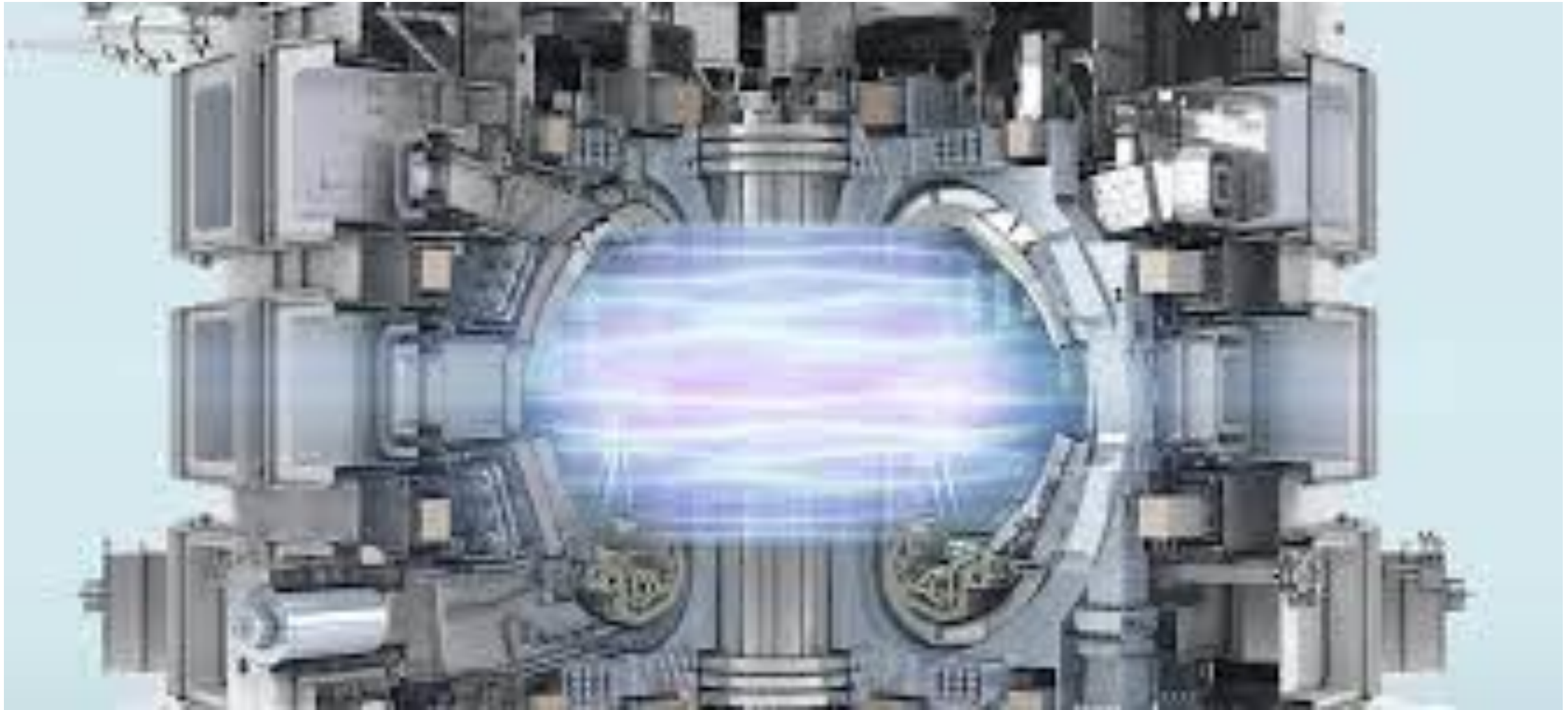
Overall Reaction:

$4\ {}^{1}H \rightarrow {}^{4}He + 2\ e^{+} + 2\ \nu_e + \gamma + 26.7\ MeV$

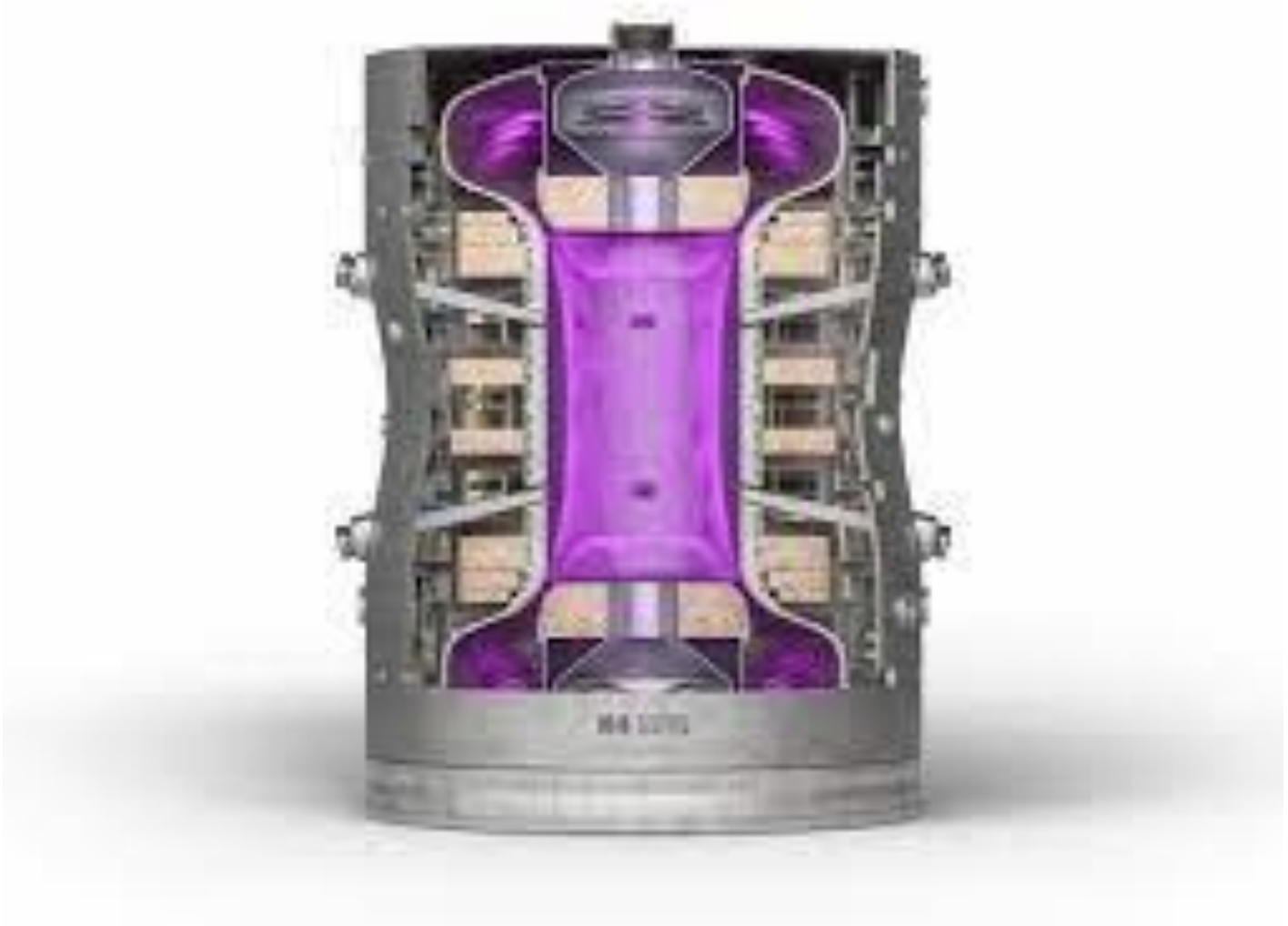
JET Tokamak



ITER



Novatron Fusion – KTH start-up

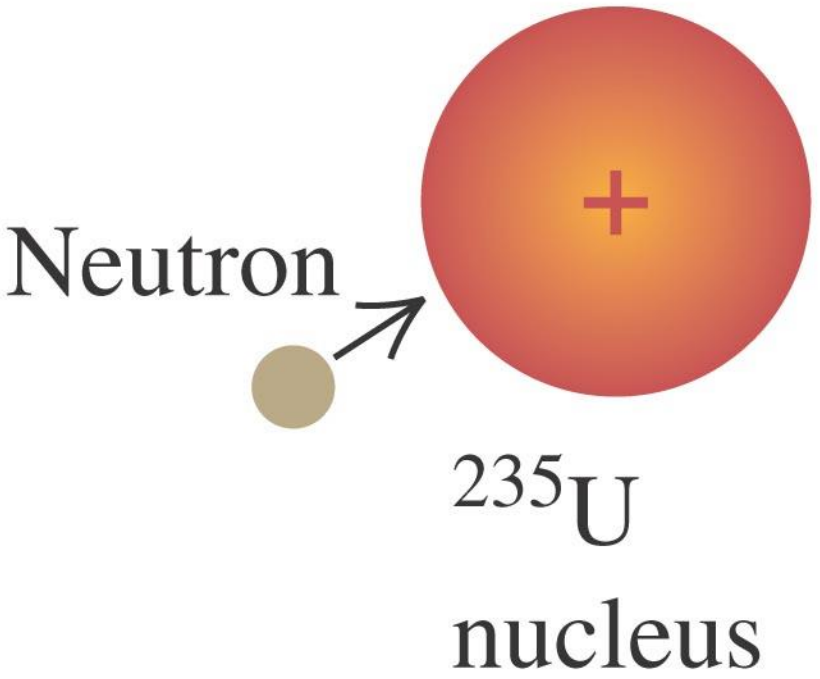


Novatron.se

Fission

Figure 43.13a

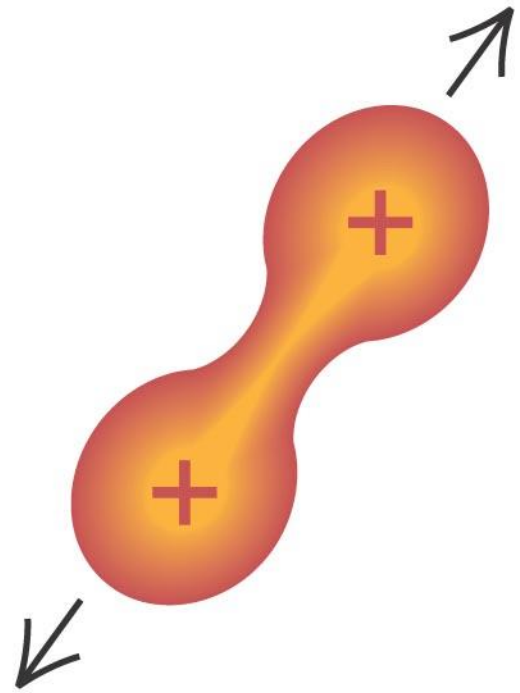
(a) A ^{235}U nucleus absorbs a neutron.



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Figure 43.13c

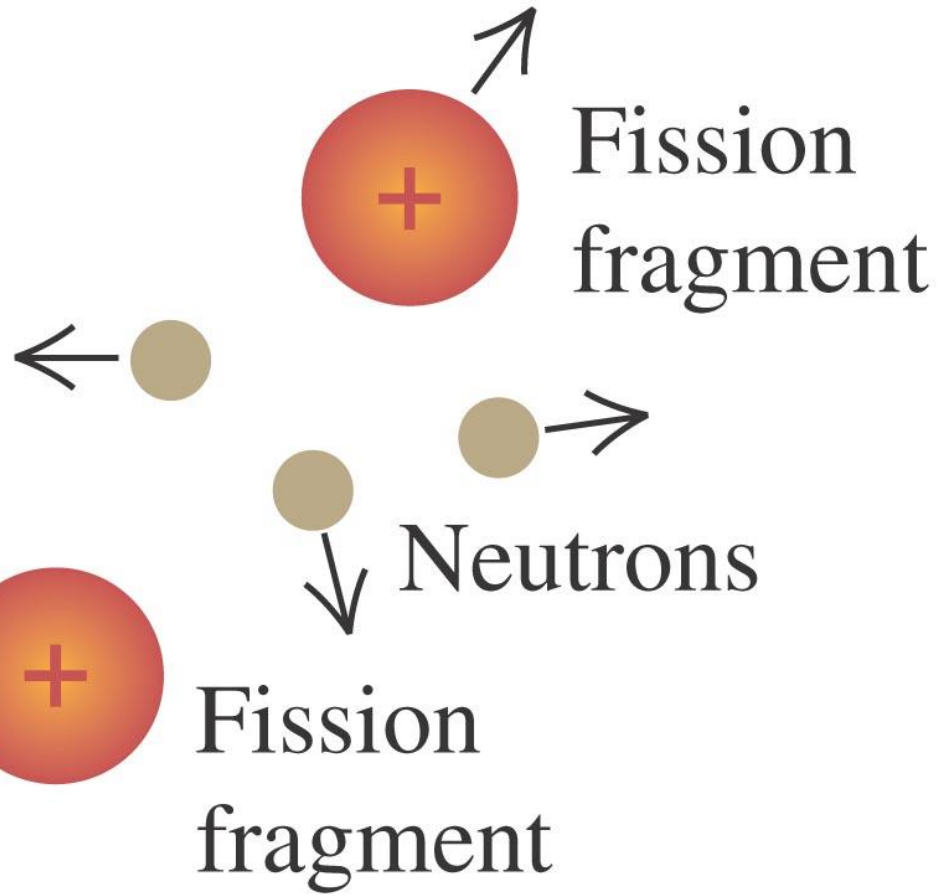
(c) A neck develops, and electric repulsion pushes the two lobes apart.



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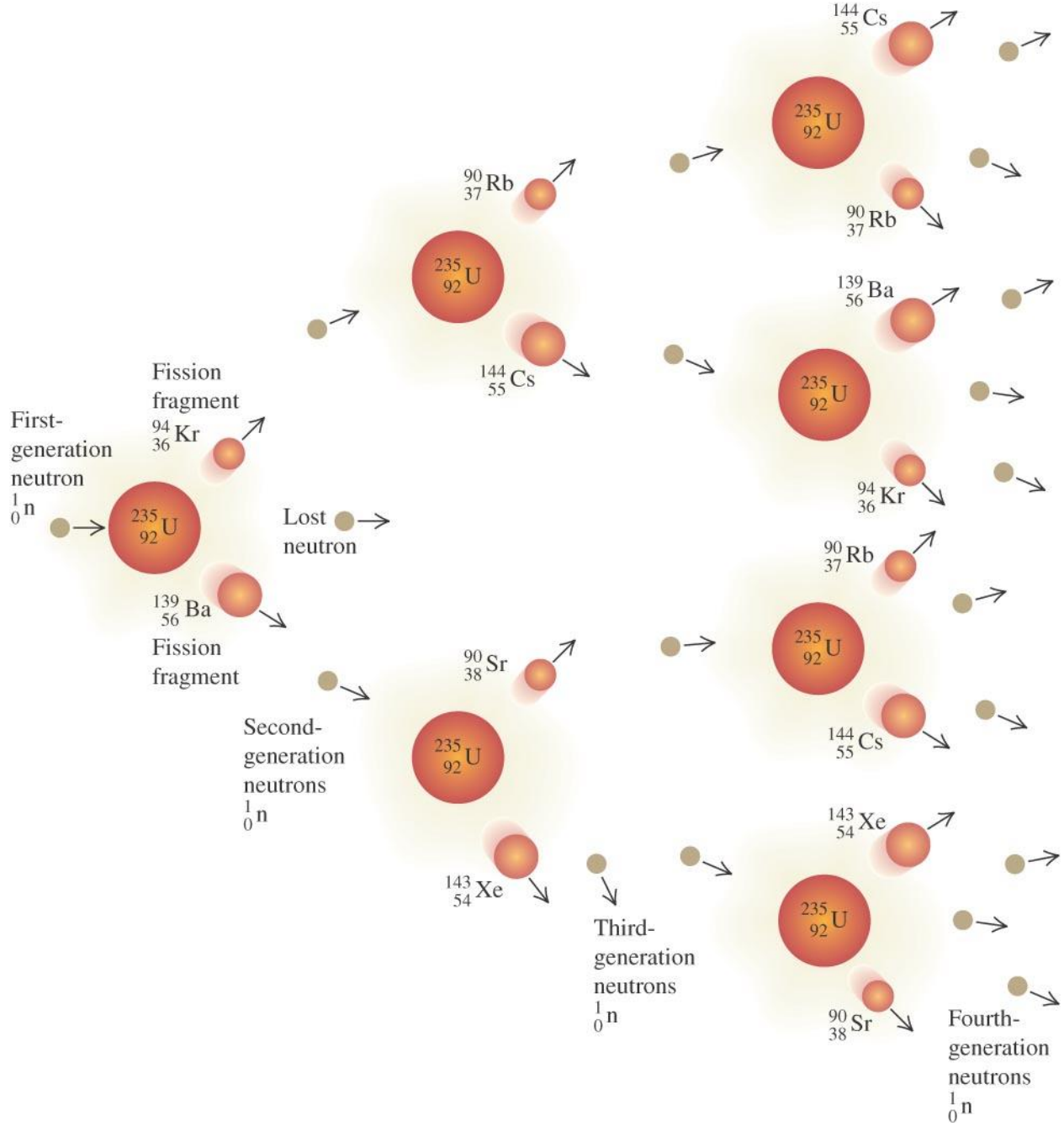
Figure 43.13e

(e) The fragments emit neutrons at the time of fission (or occasionally a few seconds later).



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Figure 43.15



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Blykalla KTH start-up för kärnkraft

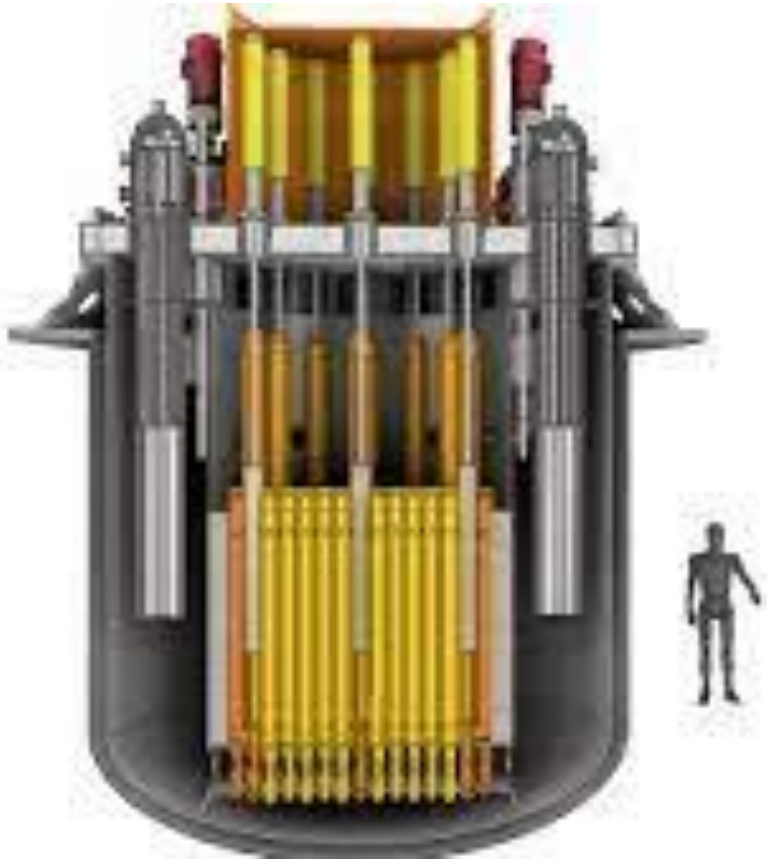
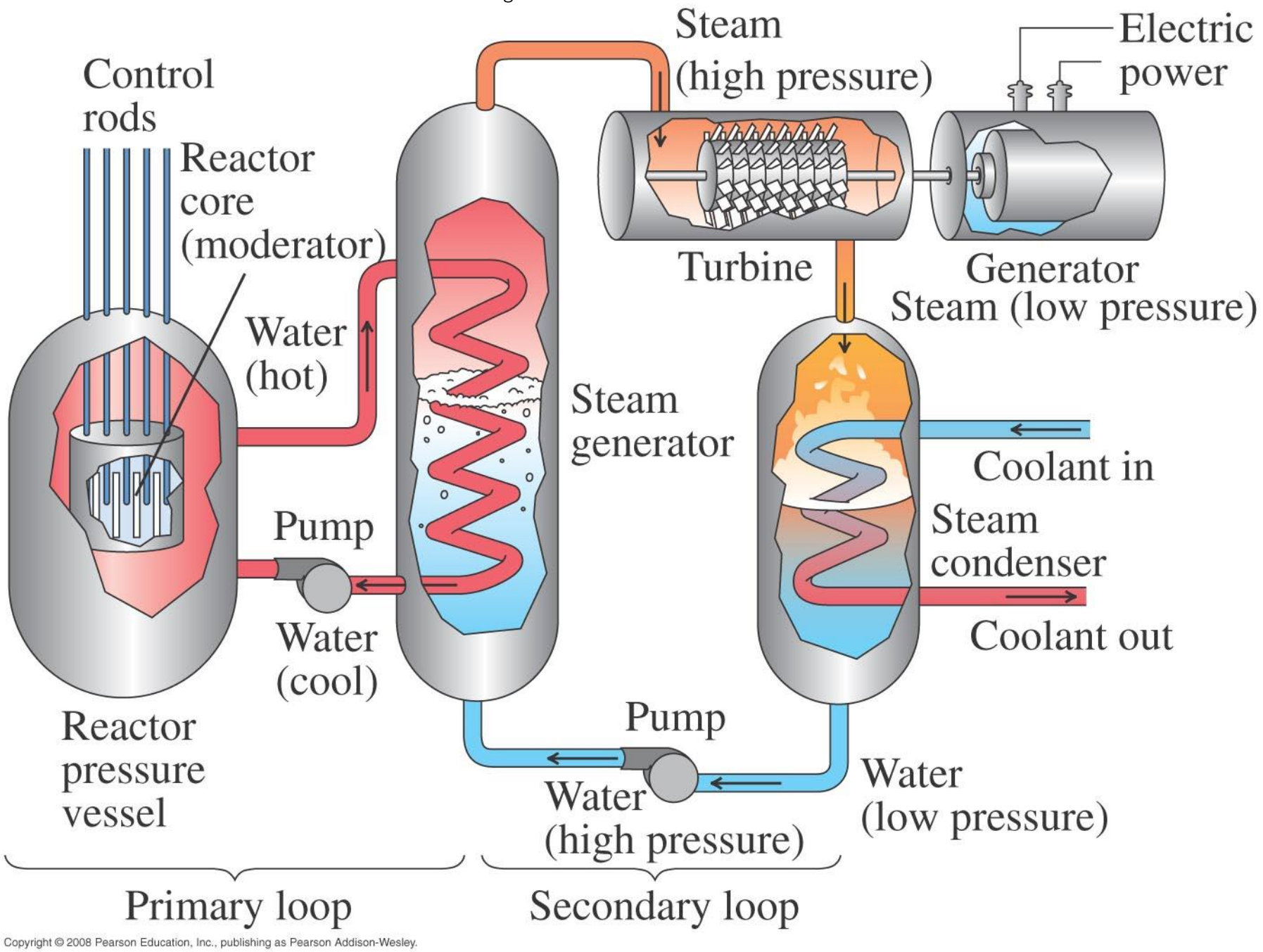




Figure 43.16



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